How-To Library

Informatica

Integrating Big Data Management 10.2.2 with a WANdisco-Enabled Cluster

© Copyright Informatica LLC 2019, 2021. Informatica, the Informatica logo, and Big Data Management are trademarks or registered trademarks of Informatica LLC in the United States and many jurisdictions throughout the world. A current list of Informatica trademarks is available on the web at https://www.informatica.com/trademarks.html.

Abstract

WANdisco fusion is a software application that replicates HDFS data among cluster nodes that are running different versions or distributions of Hadoop to prevent data loss in case a cluster node fails.

You can use Informatica Big Data Management® on Cloudera or Hortonworks clusters where WANdisco is enabled. This article describes how to configure the environment and Big Data Management on WANdisco-enabled clusters.

Supported Versions

- Big Data Management 10.2.2

Table of Contents

Overview 2
Implementation with Big Data Management 10.2.2. 2
Verify Prerequisites 2
Enable the Informatica Domain for the WANdisco-Enabled Cluster. 3
Enable the Hive Scratch Directory. 4

Overview

You can use Informatica Big Data Management® with a WANdisco-enabled cluster.

When you use Big Data Management with a supported cluster running WANdisco Fusion, HDFS data in cluster storage resources is replicated across cluster nodes.

The following table shows support across versions of Big Data Management, WANdisco, and supported clusters:

Big Data Management Version	WANdisco Version	Supported Clusters
10.2.2	2.12.4	- Cloudera 5.16 - Hortonworks 2.6.5

Implementation with Big Data Management 10.2.2

Follow the instructions in this section to implement WANdisco 2.12.4 with Big Data Management 10.2.2.

Verify Prerequisites

Verify the following prerequisites.

Informatica Prerequisites

Install Informatica Big Data Management 10.2.2 on-premises or in a cloud environment. For example, Big Data Management could be co-located in a cloud environment with a supported Hadoop cluster.

Cluster Prerequisites

Integrate WANdisco Fusion version 2.12.4 on a cluster with one of the following Hadoop distributions:

- Cloudera CDH 5.16 on RedHat 7

- Hortonworks HDP 2.6.5 on RedHat 7

Note: SUSE 11 distributions are not supported.

Enable the Informatica Domain for the WANdisco-Enabled Cluster

Copy WANdisco Fusion .jar files to all machines that host the Data Integration Service:

1. Choose the Hadoop distribution you have integrated with Big Data Management:
   - Cloudera -- Perform step 2, then steps 4-6.
   - Hortonworks -- Perform steps 3-6.
2. On the Cloudera cluster headnode, copy the following files:
   a. Browse to the following directory: `/opt/cloudera/parcels/FUSION-<version number>/lib` and copy all of the .jar files.
   b. Browse to the following directory:`/opt/cloudera/parcels/CDH-<version number>/jars` and copy the following files:
      - `hadoop-yarn-api-<version number>-cdh<version number>.jar`
      - `hadoop-yarn-common-<version number>-cdh<version number>.jar`
      - `hadoop-yarn-client-<version number>-cdh<version number>.jar`
      - `hadoop-common-<version number>-cdh<version number>.jar`
      - `hadoop-auth-<version number>-cdh<version number>.jar`
3. On the Hortonworks cluster headnode, copy the following files:
   a. Browse to the following directory: `/opt/wandisco/fusion/client/lib` and copy all of the .jar files.
   b. Browse to the following directory:`/usr/hdp/<version>/hadoop` and copy the following files:
      - `hadoop-common-<version number>-hdp<version number>.jar`
      - `hadoop-auth-<version number>-hdp<version number>.jar`
   c. Browse to the following directory: `/usr/hdp/<version>/hadoop-yarn` and copy the following files:
      - `hadoop-yarn-api-<version number>-hdp<version number>.jar`
      - `hadoop-yarn-common-<version number>-hdp<version number>.jar`
      - `hadoop-yarn-client-<version number>-hdp<version number>.jar`
4. Back up and delete the following files from each machine that hosts the Data Integration Service:
   a. Back up and delete the following files from the `<Informatica home>/services/shared/hadoop/<distribution version>/lib` directory:
      - `hadoop-yarn-api-<version number>.jar`
      - `hadoop-yarn-common-<version number>.jar`
      - `hadoop-yarn-client-<version number>.jar`
      - `hadoop-common-<version number>.jar`
      - `hadoop-auth-<version number>.jar`
   b. Back up and delete the following files from the `<Informatica home>/services/shared/spark/<Spark version>/lib` directory:
      - `hadoop-yarn-api-<version number>.jar`

- `hadoop-yarn-common-<version number>.jar`
- `hadoop-yarn-client-<version number>.jar`
- `hadoop-common-<version number>.jar`
- `hadoop-auth-<version number>.jar`

5. Paste the .jar files that you copied in steps 2 - 4 to the following locations on each machine that hosts the Data Integration Service:
   - To run jobs using the Blaze engine, paste the .jar files to `<Informatica home>/services/shared/hadoop/<distribution version>/lib`.
   - To run jobs using the Spark engine, paste the .jar files to `<Informatica home>/services/shared/spark/<Spark version>/lib`.
6. Recycle the Data Integration Service on each node.

Enable the Hive Scratch Directory

Edit the cluster configuration to enable the Data Integration Service to use the Hive scratch directory.

The Data Integration Service uses the Hive scratch directory to store temporary data about processed Hive jobs.

By default, the directory is located on the cluster at `/tmp/hive` in the HDFS location. You can specify another directory path by adding the following property in the hive-site.xml section of the HiveCCO cluster configuration:

```
hive.exec.scratchdir=<directory path>
```

If you do not set the property, the Data Integration Service uses `/tmp/hive`.

1. Optionally, set the Hive scratch directory to the preferred path.
2. Set the directory permissions to 733.
3. Set the Data Integration Service custom property ExecutionContextOptions.Spark.EnforceConfiguredHiveScratchDirUsage to TRUE.

Author

Mark Pritchard